4. The input clock to a timer module is 10 MHz, and the prescaler register TIMx_PSC is set to 999. To generate an interrupt every 0.5 seconds, what should the auto-reload value (TIMx_ARR) be?

5. The input clock to a timer is 8 MHz, and the prescaler register TIMx_PSC is set to 1023. What is the resulting timer frequency?

6. Which timer mode can be used to generate a single pulse in response to an external event?

7. Serial Communication is the act of transmitting and receiving data in parallel over a communications channel. (True/False)

8. Arrange the serial communication standards from fastest to slowest possible throughput.

9. Given that a UART COM port has been configured to operate as 4800-8-N-2:
- What is the baud rate?
- How many parity bits are there in a frame?
- How many start bits are there in a frame?
- How many stop bits are there in a frame?
- What is the data throughput in bytes/second?

10. Choose the correct answers. Incorrect answers will earn negative marks.
- SPI supports multiple masters?
- SPI only supports 8-bit transfers?
- SPI hardware controllers can also be used for transmitting [A] data?

11. You have been tasked with using a microcontroller to read and write from an SD CARD.
Indicate which pins of the SPI controller must be connected to the SD CARD.

SD Pins: SC, CMD, VSS1, VDD, CLK, VSS2, DAT0, DAT1, DAT2
Options: GPIO, MOSI, MISO, SCLK

12. A junior engineer wrote driver functions to initialize and read data from the ST LIS3DSH MEMS sensor via SPI.

You are tasked with fixing the errors in the LIS3DSH_Init and LIS3DSH_ReadXYZ functions.

The symptom: It does not work.

CODE:

```
void LIS3DSH_Init(void)
{
    HAL_StatusTypeDef res;
    uint8_t TxBuf[10] = {0};
    TxBuf[0] = 0x20 | 0x80;
    TxBuf[1] = 0x13;
    res = HAL_SPI_Transmit(&hspi1, TxBuf, 2, 100);
}

void LIS3DSH_ReadXYZ(int16_t *pData){
    HAL_StatusTypeDef res;
    uint8_t TxBuf[10] = {0};
    uint8_t RxBuf[10] = {0};
    TxBuf[0] = 0x18 ;
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_SET);
    res = HAL_SPI_Transmit(&hspi1, RxBuf, 1, 100);
    res = HAL_SPI_Receive(&hspi1, TxBuf, 3, 100);
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_RESET);
    pData[0] = RxBuf[0];
    pData[1] = RxBuf[2];
    pData[2] = RxBuf[4];
}
```

13. For I2C:

- Pull-up resistors are required on which signals?
- I2C is a full-duplex bus? (True/False)
- Clock stretching speeds up transaction throughput? (True/False)
- Which of 10, 100, 10k, or 100k Ohm would be a good initial choice for pull-up resistors?

14. For an I/O drive strength of 6mA and VDD = 3.3V, calculate the minimum pull-up resistor value (Ohms) for an I2C bus.

15. For a 010V ADC, choose the minimum resolution needed for a quantisation error of less than 1mV.

16. For a 010V 16-bit ADC, calculate the conversion result for a 1.1011V DC input signal.

17. True or False: The STM32F407 uses a Flash ADC?

18. In the STM32F407 ADC, how many clock cycles are needed for an 8-bit conversion?

19. True or False: For a SAR ADC, the conversion result is decoded instantly?

20. For a 05V 12-bit DAC, calculate the analog output value for a digital value of 0xF7 (round to 4 decimal places).

21. In a design requiring small discrete movements, one would use a [A] motor.
When switching relays on and off it is important to include a [B] in parallel with the [C].
To do efficient speed control of a DC motor, one would use an [E] to drive a [F].

22. [A] flash is used for program code.
[B] flash memory is used for mass storage.
A [C] interface has separate strobe signals for column and row addresses.
[D] information is stored as charge in a capacitor.

23. A smart power meter connects to a house's 220V line and measures current.
Requirements:
- Uses STM32F407VG
- Measures via Current Transformer (CT): +/-10mV signal <-> +/-13A current
- Calculates RMS of AC current
- Sampling rate: 1000 Hz

- Sampling via timer interrupt
- Powered from 220V mains
- Sends data via UART to wireless modem
- 3.3V supply for MCU & modem
- Flashing LED shows system status.

Complete the block diagram of this system.

24. Write an assembly function readUserPin to read the state of PA0 for the STM32F4DISCOVERY board.

C-prototype:
uint32_t readUserPin();

Notes:
- Use only input data register.
- Initialization already handled.
- Syntax checking enabled.